

Exam 3 Review

Chapters 8, 9, 10, 12

ECON3500: Econometrics and Applications

Spring 2026

Roadmap

Chapter	Topic	Key Skills
8	Nonlinear Relationships	Marginal effects, log interpretation, interactions
9	Assessing Studies	Threats to internal validity
10	Panel Data & DiD	Fixed effects, difference-in-differences
12	Instrumental Variables	2SLS, instrument validity, LATE

 **Exam 3: April 23**

~3 multi-part questions, 75 minutes, closed book. One-page formula sheet (double-sided) + calculator allowed.

Chapter 8: Nonlinear Relationships

Ch 8 Big Picture

We relax the assumption that the effect of X on Y is **constant**.

Three tools — all still **linear in parameters**, so OLS still works:

Polynomial regression — model curvature with X^2, X^3

Logarithmic transformations — model percentage changes

Interaction terms — allow the effect of X_1 to depend on X_2

Polynomial Regression

i Quadratic Model

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + u_i$$

Marginal effect of X :

$$\frac{\Delta Y}{\Delta X} \approx \beta_1 + 2\beta_2 X$$

The effect of X on Y **depends on the current value of X** .

Turning point (where the effect switches sign):

$$X^* = -\frac{\beta_1}{2\beta_2}$$

Polynomial Regression: Example

$$\widehat{TestScore} = 607.3 + 3.85 \times Income - 0.0423 \times Income^2$$

What is the marginal effect of income at $Income = 10$ (thousand)?

$$\frac{\Delta TestScore}{\Delta Income} \approx 3.85 + 2(-0.0423)(10) = 3.85 - 0.846 = 3.00$$

At $Income = 40$?

$$\frac{\Delta TestScore}{\Delta Income} \approx 3.85 + 2(-0.0423)(40) = 3.85 - 3.384 = 0.47$$

! Takeaway

You **cannot** interpret β_1 or β_2 in isolation. Always compute the **marginal effect at a specific value of X** .

The Log Interpretation Table

Specification	Model	Interpretation of
Level-level	$Y = \beta_0 + \beta_1 X$	A 1-unit $\uparrow$ in $X \rightarrow \beta_1$ unit $\uparrow$ in Y
Log-level	$\ln(Y) = \beta_0 + \beta_1 X$	A 1-unit $\uparrow$ in $X \rightarrow (100 \times \beta_1)\% \uparrow$ in Y
Level-log	$Y = \beta_0 + \beta_1 \ln(X)$	A 1% $\uparrow$ in $X \rightarrow (\beta_1/100)$ unit $\uparrow$ in Y
Log-log	$\ln(Y) = \beta_0 + \beta_1 \ln(X)$	A 1% $\uparrow$ in $X \rightarrow \beta_1 \% \uparrow$ in Y (elasticity)

 Must Know

This table is one of the **highest-value things** to have on your formula sheet. Know how to apply it.

Log Interpretation: Practice

$$\ln(\widehat{Wage}) = 1.5 + 0.08 \times Educ$$

This is log-level. One additional year of education is associated with an approximately **8% increase** in wages.

$$\widehat{Wage} = 12.5 + 3.2 \times \ln(Experience)$$

This is level-log. A 1% increase in experience is associated with a $3.2/100 = \$0.032$ increase in wages.

Interaction Terms: The Idea

The effect of X_1 on Y may **depend on the value of another variable** X_2 .

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 (X_{1i} \times X_{2i}) + u_i$$

Marginal effect of X_1 :

$$\frac{\Delta Y}{\Delta X_1} = \beta_1 + \beta_3 X_2$$

The effect of X_1 **varies with** X_2 .

Three Types of Interactions

1. Binary $\times$ Binary ($D_1 \times D_2$)

$$Y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 (D_{1i} \times D_{2i}) + u_i$$

β_3 = **additional** effect of having *both* $D_1 = 1$ and $D_2 = 1$, beyond their separate effects

2. Binary $\times$ Continuous ($D \times X$)

$$Y_i = \beta_0 + \beta_1 D_i + \beta_2 X_i + \beta_3 (D_i \times X_i) + u_i$$

Effect of X when $D = 0$: β_2

Effect of X when $D = 1$: $\beta_2 + \beta_3$

β_3 = how much the **slope** changes when $D = 1$

3. Continuous $\times$ Continuous ($X_1 \times X_2$)

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 (X_{1i} \times X_{2i}) + u_i$$

Effect of X_1 : $\beta_1 + \beta_3 X_2$ (depends on X_2)

Interaction Terms: Practice

$$\widehat{TestScore} = 700 - 1.0 \times STR - 0.5 \times PctEL - 0.02 \times (STR \times PctEL)$$

where STR = student-teacher ratio and $PctEL$ = percent English learners.

Q: What is the effect of reducing STR by 1 in a district where PctEL = 20%?

$$\frac{\Delta TestScore}{\Delta STR} = -1.0 + (-0.02)(20) = -1.0 - 0.4 = -1.4$$

Reducing STR by 1 raises test scores by **1.4 points** in a district with 20% English learners.

Q: What about in a district where PctEL = 0?

$$\frac{\Delta TestScore}{\Delta STR} = -1.0 + (-0.02)(0) = -1.0$$

Reducing STR by 1 raises test scores by **1.0 point**.

Knowledge Check

Does the interaction term suggest a **larger** or **smaller** effect of reducing class size in districts with many English learners?

Ch 8 Summary

Tool	Use When...	Key Skill
Polynomial (X^2)	Effect of X changes as X increases	Marginal effect at specific X
Logs	Think in percentages; skewed data	Log interpretation table
Interactions	Effect of X_1 depends on X_2	Marginal effect at specific X_2

Common Mistakes

- . Interpreting β_1 alone in a polynomial or interaction model
- . Forgetting to multiply by 100 (or divide by 100) for log models
- . Including the interaction but forgetting to include the **constituent terms**

Chapter 9: Assessing Studies

Ch 9 Big Picture

Chapter 9 gives a **framework** for evaluating regression studies.

Two dimensions:

Internal validity: Are the statistical inferences correct *for the population studied*?

External validity: Can the results be *generalized* to other populations and settings?

Internal vs. External Validity

Internal Validity

The statistical inferences about causal effects are valid **for the population being studied**.

External Validity

The statistical inferences can be **generalized** to other populations and settings.

Internal validity is about **getting the right answer** for your sample/population.

External validity is about **applying that answer elsewhere**.

There is no mechanical test for external validity — it requires **judgment**.

The Five Threats to Internal Validity

When we estimate $Y_i = \beta_0 + \beta_1 X_i + u_i$, internal validity requires:

$E(u_i | X_i) = 0$ — no bias

Correct standard errors — valid inference

Five threats that can violate these:

Omitted variable bias

Wrong functional form

Measurement error (errors-in-variables)

Sample selection bias

Simultaneous causality

Threat 1: Omitted Variable Bias

An omitted variable Z causes bias if:

Z is a determinant of Y (i.e., Z is in the error term), **and**

Z is correlated with X

Solution: Include the omitted variable, use proxy variables, use panel data (Ch 10), or use instrumental variables (Ch 12).

Watch Out

Adding “kitchen sink” controls can introduce its own problems — only control for variables that satisfy the OVB conditions. Bad controls can **create** bias rather than remove it.

Threat 2: Wrong Functional Form

The true relationship is nonlinear, but you estimate a linear model.

Example: The effect of income on test scores is quadratic, but you fit a line.

Solution: Use the tools from Chapter 8 — polynomials, logs, interactions. Test for nonlinearity.

Threat 3: Measurement Error

The variable you observe ($\tilde{X}$) differs from the true variable (X):

$$\tilde{X}_i = X_i + w_i$$

where w_i is the measurement error.

! Attenuation Bias (Classical Measurement Error)

If the measurement error is classical (uncorrelated with X), then OLS is biased **toward zero**.

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 \times \frac{\sigma_X^2}{\sigma_X^2 + \sigma_w^2}$$

The ratio $\frac{\sigma_X^2}{\sigma_X^2 + \sigma_w^2} < 1$, so $|\hat{\beta}_1|$ is too **small**.

Solution: Get better data, use IV (Ch 12).

Threat 4: Sample Selection Bias

The sample is not representative of the population due to a **selection process related to Y** .

Examples:

Studying wages only for employed people (employed $\neq$ random)

Studying firm performance for surviving firms (survivorship bias)

Volunteers for a study differ from non-volunteers

Solution: Use a randomly selected sample, correct for selection (Heckman correction), or be cautious about generalization.

Threat 5: Simultaneous Causality

X causes Y , but Y also causes X .

Examples:

Police and crime: More police $\rightarrow$ less crime, but more crime $\rightarrow$ more police

Class size and test scores: Low scores may prompt smaller classes

Solution: Instrumental variables (Ch 12), randomized experiments, natural experiments.

Summary Table: Threats to Internal Validity

Threat	Problem	Consequence	Solution
Omitted variable	Relevant variable left out	Bias in $\hat{\beta}_1$	Add controls, panel data, IV
Functional form	Wrong shape modeled	Bias in $\hat{\beta}_1$	Polynomials, logs, interactions
Measurement error	X measured with error	Attenuation bias (toward 0)	Better data, IV
Sample selection	Non-random sample	Bias in $\hat{\beta}_1$	Random sampling, correction
Simultaneous causality	$X \leftrightarrow Y$	Bias in $\hat{\beta}_1$	IV, experiments

 Knowledge Check

A researcher studies the effect of hospital spending on patient outcomes, but only includes patients who survived. Which threat applies?

...

Sample selection bias — the sample is conditioned on an outcome (survival) related to Y .

Causal Diagrams (DAGs): Review

A **directed acyclic graph (DAG)** maps out the causal relationships between variables.

Directed: Arrows point from cause → effect

Acyclic: No loops — causality flows forward

Graph: Nodes (variables) and edges (arrows)

! Why DAGs Matter for This Exam

DAGs give a **visual framework** for diagnosing the five threats — especially OVB, simultaneous causality, and measurement error.

Three Types of Paths



Path Type	Structure	Default	Example
Causal (front-door)	$X \rightarrow M \rightarrow Y$	Open	Education $\rightarrow$ Job $\rightarrow$ Earnings
Backdoor	$X \leftarrow Z \rightarrow Y$	Open	Class Size $\leftarrow$ Wealth $\rightarrow$ Scores
Collider	$X \rightarrow Z \leftarrow Y$	Closed	Talent $\rightarrow$ Hiring $\leftarrow$ Connections

Causal paths = what we want to estimate. Leave them open.

Backdoor paths = confounding. Must close them.

Collider paths = safely closed. Don't open them!

Confounding = OVB = Open Backdoor Path



Gray node = unobserved

Every case of **omitted variable bias** is an open backdoor path:

- Education $\leftarrow$ Ability $\rightarrow$ Earnings
- Ability is unobserved $\rightarrow$ can't control for it
- OLS picks up causal effect + confounding bias

...

$$\text{Association} = \underbrace{\text{Causal Effect}}_{\text{front-door}} + \underbrace{\text{Bias}}_{\text{open backdoor}}$$

What “Controlling For” Does

Controlling for a variable **blocks all paths** through it. This can be good or bad:

Path Type	Default	Controlling...	Result
Backdoor ($X \leftarrow Z \rightarrow Y$)	Open	Closes it	Removes bias ✓
Collider ($X \rightarrow Z \leftarrow Y$)	Closed	Opens it	Creates bias ✗
Mediator ($X \rightarrow Z \rightarrow Y$)	Open	Closes it	Blocks causal effect ✗

 Bad Controls

Don't control for mediators — you'll block the effect you're trying to measure

Don't control for colliders — you'll introduce bias where there was none

Do control for confounders — they create open backdoor paths (= OVB)

DAG Practice



Knowledge Check

You want to estimate the causal effect of Health Insurance on Health. Income affects both.

1. List all paths from Insurance to Health
2. Classify each as causal or backdoor
3. What should you control for?

Paths:

Insurance $\rightarrow$ Health (causal — leave open)

Insurance $\leftarrow$ Income $\rightarrow$ Health (backdoor — **control for Income**)

After controlling for Income, the only open path is causal. Effect is identified.

Collider Bias: Practice



Knowledge Check

You want to know if Talent affects Connections. Both cause Job Hiring.

Should you study this relationship **among hired workers only**?

No! Conditioning on Job Hiring (the collider) **opens** the path between Talent and Connections.

Among the hired, low Talent becomes correlated with high Connections — a spurious association.

This is **collider bias** (a.k.a. selection bias).

DAGs and Our Econometric Toolkit

DAGs help us see **why** each method works:

Problem (in DAG terms)	Method	How it helps
Open backdoor via observed confounder	Add control variable	Closes the backdoor path
Open backdoor via time-invariant unobserved confounder	Fixed effects (Ch 10)	Absorbs all α_i variables
Open backdoor via any unobserved confounder	IV / 2SLS (Ch 12)	Isolates exogenous variation in X
Open backdoor; treatment + control available	DiD (Ch 10)	Parallel trends closes the path
No backdoor paths (by design)	RCT	Randomization severs arrows into X

 Cross-Chapter Connection

When a problem asks “which method would address this threat?” — draw the DAG first. The answer becomes visual.

Ch 9: What to Focus On

Be able to **name and define** each of the five threats

Given a research scenario, **identify which threat(s)** apply

Know the **direction of bias** when possible (especially OVB and attenuation bias)

Know at least one **solution** for each threat

Draw and interpret simple DAGs — identify causal vs. backdoor vs. collider paths

Know what to **control for** and what **not to control for** (mediators, colliders)

Explain why confounding = OVB = open backdoor path

Chapter 10: Panel Data & DiD

Ch 10 Big Picture

Panel data = same units observed over multiple time periods.

Two powerful tools:

Fixed effects — control for all time-invariant unobserved characteristics

Difference-in-differences — estimate causal effects by comparing changes over time between treatment and control groups

Types of Data

Type	Units	Time Periods	Example
Cross-sectional	Many	One	Survey of 1000 workers in 2025
Time series	One	Many	US GDP, 1950–2025
Panel	Many	Many	50 states, 2000–2025

Panel vs. Pooled Cross-Sections

A **panel** tracks the **same units** over time (Bob in 2020 and Bob in 2021).

A **pooled cross-section** draws **different samples** each period (different people each year).

Why Panel Data?

The key advantage: we can control for **unobserved, time-invariant** characteristics.

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \alpha_i + u_{it}$$

α_i = **entity fixed effect** — captures everything about unit i that doesn't change over time

Examples: ability, geography, institutional culture, baseline preferences

! Key Insight

Fixed effects eliminate OVB from **any** time-invariant omitted variable — even ones you can't observe or measure.

Entity Fixed Effects: Mechanics

Approach 1: First-difference estimator

Take the difference between two periods:

$$Y_{it} - Y_{i,t-1} = \beta_1 (X_{it} - X_{i,t-1}) + (u_{it} - u_{i,t-1})$$

The fixed effect α_i **drops out** because $\alpha_i - \alpha_i = 0$.

Approach 2: Entity dummies (LSDV)

Include a dummy variable for each entity:

$$Y_{it} = \beta_1 X_{it} + \gamma_1 D_{1i} + \gamma_2 D_{2i} + \cdots + u_{it}$$

Approach 3: Demeaning

Subtract the entity mean: $\tilde{Y}_{it} = Y_{it} - \bar{Y}_i$, then regress $\tilde{Y}$ on $\tilde{X}$.

All three approaches give the **same** $\hat{\beta}_1$.

Time Fixed Effects

$$Y_{it} = \beta_1 X_{it} + \alpha_i + \lambda_t + u_{it}$$

λ_t = **time fixed effect** — captures anything affecting all units in period t

Examples: macroeconomic shocks, federal policy changes, technological trends

Entity FE + Time FE together control for:

All time-invariant unit characteristics (α_i)

All unit-invariant time shocks (λ_t)

Limitation

Fixed effects **cannot** control for omitted variables that change over time *and* vary across units. For that, you may need IV (Ch 12) or DiD.

Difference-in-Differences: Setup

DiD estimates a causal effect by comparing **changes over time** between a **treatment group** and a **control group**.

	Before	After	Change
Treatment	$\bar{Y}_{T,before}$	$\bar{Y}_{T,after}$	$\Delta \bar{Y}_T$
Control	$\bar{Y}_{C,before}$	$\bar{Y}_{C,after}$	$\Delta \bar{Y}_C$

DiD estimate:

$$\begin{aligned}\hat{\delta}_{DiD} &= (\bar{Y}_{T,after} - \bar{Y}_{T,before}) - (\bar{Y}_{C,after} - \bar{Y}_{C,before}) \\ &= \Delta \bar{Y}_T - \Delta \bar{Y}_C\end{aligned}$$

DiD as a Regression

$$Y_{it} = \beta_0 + \beta_1 \text{Treat}_i + \beta_2 \text{After}_t + \beta_3 (\text{Treat}_i \times \text{After}_t) + u_{it}$$

Coefficient	Meaning
β_0	Control group mean, before
β_1	Difference between treatment and control, before
β_2	Change over time for control group
β_3	DiD estimate = causal effect of treatment

 The DiD Coefficient

β_3 is the coefficient on the **interaction term** $\text{Treat} \times \text{After}$. This is the causal effect of interest.

The Parallel Trends Assumption

Parallel Trends

In the **absence of treatment**, the treatment and control groups would have followed the **same trend** over time.

This is the **key identifying assumption** of DiD.

It is **not directly testable** — we never observe what would have happened to the treatment group without treatment.

We can check **pre-treatment trends** — if they are parallel before treatment, it is more plausible they would have remained parallel.

If Parallel Trends Fails

The DiD estimate is **biased**. Any pre-existing divergence between groups gets picked up as a “treatment effect.”

DiD Calculation: Practice

Knowledge Check

A state raises its minimum wage in 2020. You have average teen employment rates:

	2019	2021
Treatment state	50%	47%
Control state	53%	52%

Calculate the DiD estimate of the minimum wage effect on teen employment.

...

Step 1: Change in treatment: $47 - 50 = -3$ pp

Step 2: Change in control: $52 - 53 = -1$ pp

Step 3: DiD = $(-3) - (-1) = -2$ pp

The minimum wage increase is associated with a **2 percentage point decrease** in teen employment.

Ch 10: What to Focus On

Distinguish cross-sectional, time-series, and panel data

Explain what **entity fixed effects** control for and what they **cannot** control for

Know how the **first-difference estimator** eliminates fixed effects (the algebra)

Calculate a DiD estimate from a 2x2 table

State the **parallel trends assumption** and explain why it matters

Know which coefficient in the regression is the DiD estimate (β_3 , the interaction)

Standard Errors in Panel Data

With panel data, observations within the same entity are likely correlated over time. Use **clustered standard errors** (cluster at the entity level) — otherwise SEs are too small and you over-reject.

Chapter 12: Instrumental Variables

Ch 12 Big Picture

When $E(u_i | X_i) \neq 0$ (endogeneity), OLS is **biased and inconsistent**.

Instrumental variables (IV) solve this by finding a variable Z that:

Affects X (so we can use it to predict X)

But has no direct relationship with Y except through X

Think of IV as **isolating** the “good” variation in X — the part unrelated to the error term.

Why Do We Need IV?

Three threats produce endogeneity ($E(u_i | X_i) \neq 0$):

Threat	Example
Omitted variable bias	Ability biases returns to schooling
Simultaneous causality	Police $\leftrightarrow$ crime
Measurement error	Self-reported income

When we cannot fix these with controls, panel data, or experiments, we turn to **instrumental variables**.

The Three Conditions for a Valid Instrument

An instrument Z must satisfy:

i 1. Relevance

Z is correlated with X : $\text{corr}(Z_i, X_i) \neq 0$

Z must actually affect X . Testable with the first-stage regression.

i 2. Exogeneity (Independence)

Z is uncorrelated with the error term: $\text{corr}(Z_i, u_i) = 0$

Z must be “as good as randomly assigned.” **Not directly testable** — requires reasoning.

i 3. Exclusion Restriction

Z affects Y **only through** X , not directly.

There is no direct causal path from Z to Y that bypasses X . **Not directly testable.**

Classic Example: Returns to Education

Problem: Estimate the causal effect of schooling on wages.

$$\ln(wage_i) = \beta_0 + \beta_1 schooling_i + u_i$$

OVB: Ability is in u_i and correlated with schooling $\rightarrow \hat{\beta}_1$ is biased upward.

Proposed instrument: Quarter of birth (Z)

Relevance: Compulsory schooling laws mean people born earlier in the year can drop out sooner $\rightarrow$ less schooling. ✓

Exogeneity: Quarter of birth is essentially random. ✓ (plausibly)

Exclusion: Quarter of birth doesn't directly affect wages — only through schooling. ✓ (debatable)

Caution

Exclusion restriction is always **arguable**, not provable. This is where most debates about IV studies focus.

Two-Stage Least Squares (2SLS)

Stage 1: First-Stage Regression

Regress X on Z (and any controls W):

$$X_i = \pi_0 + \pi_1 Z_i + \pi_2 W_i + v_i$$

Save the predicted values $\hat{X}_i$.

Stage 2: Second-Stage Regression

Regress Y on $\hat{X}$ (and any controls W):

$$Y_i = \beta_0 + \beta_1 \hat{X}_i + \beta_2 W_i + u_i$$

Why this works: $\hat{X}_i$ contains only the variation in X that comes from Z . Since Z is exogenous, $\hat{X}_i$ is uncorrelated with u_i .

Important

Use a proper IV/2SLS command (e.g., `ivregress 2sls` in Stata). Running two separate regressions gives **wrong standard errors**.

The Wald Estimator (Binary Instrument)

When the instrument Z is **binary** (0 or 1), the IV estimate simplifies:

$$\beta_1^{IV} = \frac{\bar{Y}_{Z=1} - \bar{Y}_{Z=0}}{\bar{X}_{Z=1} - \bar{X}_{Z=0}} = \frac{\text{Reduced form}}{\text{First stage}}$$

Piece	Regression	Meaning
Reduced form	Y on Z	Total effect of Z on Y (includes the path through X)
First stage	X on Z	Effect of Z on X
IV estimate	Ratio	Causal effect of X on Y , scaled by how much Z moves X

 Intuition

The reduced form tells you how much Z moves Y . The first stage tells you how much Z moves X . Dividing gives you how much X moves Y .

Weak Instruments

If Z is only **weakly correlated** with X , the IV estimator has serious problems:

Large standard errors

Biased (toward OLS) in finite samples

Unreliable hypothesis tests

! The $F > 10$ Rule of Thumb

Test instrument relevance with the **first-stage F-statistic**.

$F > 10$: Instrument is sufficiently strong ✓

$F < 10$: Weak instrument — IV estimates are unreliable ✗

This is a **necessary condition** you should always check.

How to get the F-statistic: Run the first-stage regression and look at the F-statistic on the excluded instrument(s).

LATE: What Does IV Actually Estimate?

i Local Average Treatment Effect (LATE)

IV estimates the causal effect for **compliers** — the subpopulation whose treatment status is changed by the instrument.

Four types of individuals:

Type	Behavior
Compliers	Do what the instrument “tells” them (take treatment when $Z = 1$, don’t when $Z = 0$)
Always-takers	Take treatment regardless of Z
Never-takers	Never take treatment regardless of Z
Defiers	Do the opposite of what Z “tells” them

IV estimates the effect **only for compliers** — not for the whole population.

This is a **local** estimate, not a global one.

The IV estimate may differ from the ATE if compliers are not representative — think carefully about **who the estimate applies to**.

Evaluating Instruments: Practice

Knowledge Check

A researcher wants to estimate the effect of smoking on health. She proposes using **cigarette taxes** as an instrument.

Evaluate each condition:

...

1. Relevance: Higher taxes → higher prices → less smoking. ✓ (testable)

...

2. Exogeneity: Are cigarette taxes correlated with unobserved determinants of health? Possibly — states with higher taxes may also have better health infrastructure. ⚠️

...

3. Exclusion: Do cigarette taxes affect health only through smoking? Maybe not — tax revenue could fund health programs. ⚠️

...

Verdict: Relevance is fine, but exogeneity and exclusion are **debatable**. The researcher would need to argue carefully and possibly add controls.

Ch 12 Summary

Concept	Key Point
Endogeneity	$E(u_i X_i) \neq 0 \rightarrow$ OLS is biased
Valid instrument	Relevant + exogenous + exclusion restriction
2SLS	First stage predicts X ; second stage uses $\hat{X}$
Weak instruments	$F < 10$ in first stage $\rightarrow$ unreliable
LATE	IV estimates effect for compliers only

Common Mistakes

- . Claiming an instrument is valid without arguing exclusion restriction
- . Running 2SLS as two separate regressions (wrong SEs)
- . Ignoring the weak instrument problem
- . Interpreting IV as the ATE instead of the LATE

Putting It All Together

Connections Across Chapters

These chapters are deeply linked:

Ch 8 → Ch 9: Wrong functional form (Ch 9) is solved by nonlinear models (Ch 8)

Ch 9 → Ch 10: Omitted variable bias (Ch 9) can be addressed by fixed effects (Ch 10) when the omitted variable is time-invariant

Ch 9 → Ch 12: OVB, simultaneous causality, and measurement error (Ch 9) are all forms of endogeneity addressed by IV (Ch 12)

Ch 10 + Ch 12: IV can be combined with panel data/fixed effects when endogeneity remains after controlling for fixed effects

Decision Tree: Which Tool?

You suspect your OLS estimate is biased. What do you do?

If the problem is...	Try...
Wrong functional form	Polynomials, logs, interactions (Ch 8)
Time-invariant omitted variable + panel data available	Entity fixed effects (Ch 10)
Common time shock	Time fixed effects (Ch 10)
Clear before/after treatment with a control group	DiD (Ch 10)
Endogeneity remains (OVB, simultaneity, meas. error)	Instrumental variables (Ch 12)
Non-random sample	Address selection, or be cautious

What to Bring to the Exam

! Formula Sheet Suggestions

Your double-sided formula sheet should include:

Log interpretation table (Ch 8) — you will almost certainly need this

Marginal effect formulas for polynomials and interactions (Ch 8)

Five threats to internal validity (Ch 9)

DiD formula and regression setup (Ch 10)

Three conditions for a valid instrument (Ch 12)

2SLS steps (Ch 12)

$F > 10$ rule (Ch 12)

Exam Logistics

Exam 3 Details

Date: Thursday, April 23

Format: ~3 multi-part questions, 75 minutes

Closed book — one-page formula sheet (double-sided), calculator allowed

No Stata component

Practice resources:

Problem Sets 4 and 5

Practice exam (linked from course website)

Lecture slides and knowledge checks

 **Study Strategy**

- . Redo the knowledge checks from each chapter's slides
- . Work through the practice exam under timed conditions
- . Make your formula sheet *before* studying — the act of choosing what goes on it is studying